27.12MHz GaN Bi-directional Resonant Power Converter

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***Abstract*—Resonant dc-dc converter regularly use Schottky diodes for high frequency rectification. Si Schottky diodes have relatively low voltage ratings and present reverse-recovery type losses when driven hard enough. SiC Schottky diodes can operate at high frequencies, but incur higher conduction losses. This paper presents a bidirectional dc-dc resonant converter having two ground referenced switches. The prototype converter delivers 400 W between a source of 170 V and a load of 50 V while delivering 250 W in the opposite direction. The converter presented herein can potentially be used in applications such as wireless power transfer, fuel cell and battery applications, bus converters, etc.**

I. Introduction

A principal method of improving power density, transient response, and realizing fully integrated of state-of-art power converters is to increase switching frequencies. Previous work has demonstrated successful dc-ac, ac-dc, and dc-dc power conversion at switching frequencies greater than 10 MHz, and even beyond 100 MHz [1]–[4] through the use of resonant topologies. The common structure of a resonant dc-dc converter in the above mentioned work consists of an resonant inversion stage, an impedance transformation stage, and a resonant rectification stage as shown in Fig. 1.

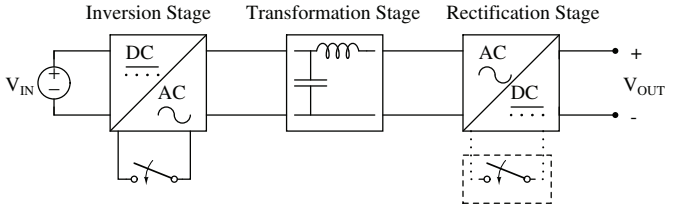


Fig. 1: Structure of the power stage of a resonant dc-dc converter. The converter comprises an inversion stage, a transformation stage, and a rectification stage.

In order to operate with high switching frequencies (>10 MHz), Schottky diodes are normally used in the rectifier stage of both hard- and soft-switching converters. However, leakage current in Silicon Schottky diodes is significant and limit the voltage rating of available devices. Further, diodes that operate at mid to high power levels (voltage ratings of 100s of volts and current ratings of several amps), the switching time listed in the manufacturer's datasheet is in the order of 10s of nano-seconds [5]. These switching times are comparable to the switching frequencies of converters operating at greater than 10 MHz which are the focus of this study. Evidence shows that some Si Schottky diodes when driven hard enough still present reverse-recovery-like losses at switching frequency above 1 MHz [6], [7]. Available SiC Schottky devices have voltage ratings much higher than their Si counterparts and can reach 3 kV or more, making them an attractive option in higher voltage applications. But these devices have a higher forward drops that can be about 3-4 times higher than Si devices. Higher forward voltage drop bring higher conduction losses which limits efficiency particularly in lower voltage converter applications (e.g. point of load converters, etc.). Moreover, it has been shown that SiC Schottky devices can present higher frequency dependent losses than expected when operating under high dv/dt conditions [8].

A common way to reduce conduction losses in rectifier circuits at high frequencies is through synchronous rectification [7], [9]–[12] in which MOSFETs are used instead of diodes in the circuit. In such circuits, the conduction losses are reduced when the MOSFET replacing the diode meets the condition that $R_{ds,ON} I^2_{rec,RMS} < V_F I_{rec}$ where $R_{ds,on}$ is the ON resistance of the MOSFET and V_F is the forward drop of the diode replaced. However, one has to consider the impact in performance, complexity and cost of the added circuitry necessary to synchronously drive the additional MOSFETs. This is especially important as frequencies approach 10 MHz and beyond. An additional benefit of synchronous rectification is that it enables bidirectional power transfer [10], [13]–[17]. Moreover, with recent advances in the development of Gallium Nitride (GaN) Field Effect Transistor (FET), Wide Bandgap (WB) Devices having lower on-state resistance and low parasitic capacitances can further benefit synchronous rectification in mid power level applications.

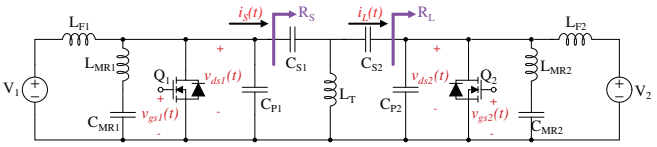
This paper presents the design and implementation of a 27.12 MHz bidirectional resonant dc-dc converter with synchronous rectification using GaN FETs based on a Φ_2 inverter/rectifier [1]–[3]. The prototype converter delivers 400 W between a source of 170 V and a load of 50 V while delivering 250 W in the opposite direction. The schematic of the circuit implemented is shown in Fig. 2a. Compared to conventional PWM synchronous dc-dc converters (i.e. PWM synchronous buck/boost converter), and resonant synchronous dc-dc converters (i.e LLC resonant converter with half-bridge synchronous rectifier [7]), this implementation utilizes two ground-referenced switches, avoiding the complexity of designing a high frequency gate drive circuit. Bidirectional power flow is therefore simple to realize in the converter described here. Such converter can find application in wireless

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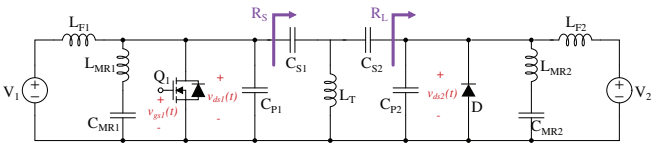
power transfer systems, modular circuits combined to produce a higher voltage ($\geq$ 3 kV) system [13], fuel cell and battery application [14]–[17], bus converters, among others. Section II introduce the circuit topology. Section III illustrates the design procedure of the prototype converter. Experimental results of the working prototype are presented in Section IV. Section V concludes the paper.

II. Circuit Topology

Referring to the circuit schematic of a single-stage resonant dc-dc converter (shown in Fig. 1), the prototype converter implemented for this study consists of a Φ_2 inverter [1]–[3], [9], [18], followed by a high-pass T matching network connected to a Φ_2 rectifier. In a synchronous rectification converter capable of bidirectional power flow, an active switch replaces the diode in the otherwise unidirectional dc-dc converter (see Fig. 2b [18]). As virtually all power is delivered only at the fundamental of the switching frequency, the converter stage serving as a synchronous resonant rectifier is designed to have a resistive impedance at the switching frequency of operation (that is, the fundamental of voltage V_{REC} and the fundamental of the drive current I_{REC} are in phase). The matching network transforms the resistance of the rectifier to the inverter's design value [2].



(a) Bidirectional $\Phi 2$ dc-dc converter



(b) Unidirectional $\Phi 2$ dc-dc converter

Fig. 2: $\Phi 2$ dc-dc converter

The high-pass T matching network used here is an alternative to the networks used in [9], [10], [18]. As shown in Fig. 2a, the T network makes the entire circuit topologically symmetric and easy to operate bidirectionally. In Fig. 2a, C_{S1} and C_{S2} are resonant components of relative small value while at the same time, provide dc blocking [1], [2]. These small value capacitors allow for faster transient response and can be potentially incorporated into the PCB. Moreover, C_{S1} and C_{S2} can be split in two, with one capacitor placed in the return path to obtain capacitive isolation as demonstrated in [10], [19]. Also, L_T can be replaced with a transformer as an alternative way of achieving galvanic isolation.

The Φ_2 inverter/rectifier resonantly transfers energy between dc and ac. It achieves Zero Voltage Switching (ZVS) which nearly eliminates switching loss, and also operates with zero dv/dt [1], [2], a characteristic common to the Class E Power amplifier [20]. The voltage waveform at the drain node of the switch is nearly trapezoidal with an average voltage equal to the dc voltage supplied. When operating in forward direction, V_1 is the input voltage and V_2 is the output voltage, and vice versa.

III. Prototype Bidirectional Converter Design

The design of the bidirectional dc-dc converter presented in this paper followed 3 steps: *a)* the design of the inverter and rectifier; *b)* design of the matching network; and *c)* gate drive implementation. The voltage stress across the switch in a Φ_2 inverter/rectifier is about two times of the dc input voltage. The design in this paper assumes an input voltage of 170 V based on the rectified ac line voltage in the United States. Two 650 V GS66508T GaN FETs from GaN Systems were chosen allowing the maximum input voltage of the Φ_2 inverter/rectifier to be approximately 300 V. Here we choose V_{IN}=170 V and V_{OUT}=50 V for the single stage prototype. The stacked stage topology of [21] can be used to extend the input voltage to the universal wall voltage range. Moreover, the bidirectionallity of the dc-dc converter can be applied to the high frequency regulating dc-dc converter in [21]. The design specifications for the converter are listed in Table I.

TABLE I: Target specifications of prototype dc-dc converter

Parameter	Value	Units
f_s	27.12	MHz
V_{IN}	170	V
V_{OUT}	50	V
P_{OUT}	400	W

A. Inverter and Rectifier Stage

The guidelines for selecting component values for the inverter are based on an iterative impedance tuning method detailed in [1]. While the design herein has no L_S in series with the load of the inverter, the tuning process remains the same as described in [1], [2].

Assuming the transformation stage is lossless for now (later we will demonstrate the impact of using lossy components), the power at the drain node of Q_1 (P_{INV}) and the power at the drain node of Q_2 (P_{REC}) are equal. With an estimated efficiency η_{REC} of the rectifier section,

$$P_{INV} = P_{REC} = \frac{P_{OUT}}{\eta_{REC}}. \tag{1}$$

Assuming all the power is delivered at the fundamental, the relationships between P_{INV} and P_{REC} and the source impedance, R_S, and the load impedance, R_L, of the matching network are calculated by

$$\begin{aligned} R_S &= \frac{(\frac{4}{\pi}V_{IN})^2}{2P_{INV}} \\ R_L &= \frac{(\frac{4}{\pi}V_{OUT})^2}{2P_{REC}} \end{aligned} \tag{2}$$

For the prototype converter, R_S = $(\frac{4}{\pi}170\text{ V})^2/(2*400\text{ W})$ = 58 Ω, R_L = $(\frac{4}{\pi}50\text{ V})^2/(2*400\text{ W})$ = 5 Ω. With these two resistance values, the resonant components in the inverter and rectifier are selected and listed in Table II using the principles in [1]. The effective C_{P1} and C_{P2} in Fig. 2a include the nonlinear C_{OSS} of Q_1 and Q_2 in addition to the listed values, respectively.

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TABLE II: Componenets of the prototype converter

Component	Designed	Actual	Description
		170V inverter/rectifier	
Q_1	-	-	GS66508T 650V eGaN FET
L_1	104nH	101nH	T68-0 10 turns 2*AWG18
L_{MR1}	229.5nH	207nH	T68-0 15 turns AWG18
C_{MR1}	37.5pF	18pF×2+1pF×2	1500V Ceramic NP0
C_{P1}	57pF	30pF+27pF	500V Ceramic NP0
C_{IN}	-	1uF×4	250V Ceramic X7R
		Matching network	
L_T	109.85nH	108nH	T68-0 10 turns AWG14
C_S	129.88pF	100pF+30pF	500V Ceramic NP0
C_L	279.45pF	100pF×2+68pF	500V Ceramic NP0
		50V rectifier/inverter	
Q_2	-	-	GS66508T 650V eGaN FET
L_2	46nH	46nH	T68-0 6 turns 3*AWG18
L_{MR2}	143.5nH	136.8nH	T68-0 11 turns AWG18
C_{MR2}	60pF	30pF×2+1pF×2	1500V Ceramic NP0
C_{P2}	10pF	0	-
C_{OUT}	-	1uF×4	250V Ceramic X7R

B. Transformation Stage

The transformation stage can be implemented using either transformers or matching networks [1], [2], [18]. Utilizing a combination of these networks can bring benefits such as large conversion ratios and a wide range of input/output voltage and power levels [22], as well as active impedance control capability [23]. A topology that eliminates the transformation stage by stacking multiple inverter/rectifier stages to achieve large voltage gain has been demonstrated in [19]. The transformation stage used here is a simple high-pass T matching network. Values of R_S and R_L were obtained as described above using Equation (2). The Q of the high-pass T matching network is a design parameter with a minimum value,

$$Q_{min} = \sqrt{\frac{R_S}{R_L} - 1} \tag{3}$$

In this design, the minimum value of Q, $Q_{\text{min}} = \sqrt{58/5 - 1} = 3.26$. We selected $Q = 4.2$, and the corresponding component values are listed in Table II. The small signal impedance looking into the drain of Q_2 is measured through an Agilent Technologies E5061B Network Analyzer and shown in Fig. 3. The figure shows that the components with parasitics on the PCB are tuned to match the designed values.

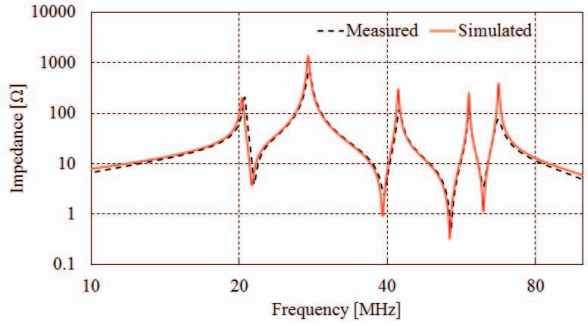


Fig. 3: Small signal impedance looking into the drain node of Q_2

The matching network sees an effective resistance on each side at the fundamental of the switching frequency. The fundamental component of the voltage on the left side $v_{ds1}(t)$, termed V_{ds11}, is in phase with the fundamental component of the current $i_S(t)$ flowing into the left of the matching network, termed as I_{S1} . The same with V_{ds21} and I_{L1}. However, the phase shift Θ between the fundamental components V_{ds11} and V_{ds21} is determined by the components of the transformation stage. The phase shift Θ plays an important role in setting the delay that is necessary to achieve between the gate drive signals of the rectifier and the inverter switch. If using a transformer (assumed ideal), Θ is 0, and Q_1 and Q_2 would turn on at the same time. For the T matching network used here, the phase shift can be found from the transfer function of the simplfied circuit shown in Fig 4. With the selected values of C_{S1}, L_T, and C_{S2} in Table II, V_{ds21} is leading V_{ds11} by 115° at 27.12 MHz.

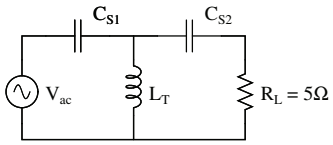


Fig. 4: Equivalent circuit

C. Gate Drive Implementation

The gate drive signals $v_{gs1}(t)$ and $v_{gs2}(t)$ determine the timing and duration of the turn on period of each switch. The duty cycles of $v_{gs1}(t)$ and $v_{gs2}(t)$ are determined during the tuning process to achieve ZVS in Q_1 and Q_2. This was accomplished though simulation. During the design, the 170 V inverter was loaded with a 58 Ω resistance. During the rectifier tuning process, the circuit was driven by a sinusoidal current source with an amplitude of 12.5 A. This operating condition results in an effective input resistance of 5 Ω at the fundamental frequency. As part of an iterative tuning process, the duty cycle of the 170 V inverter was selected to be 30%, while the duty cycle of 50 V rectifier is 44%, as shown in Fig. 5. The time delay between the drive signal of Q_2 and the drive signal of Q_1 is then determined by the phase shift Θ and the difference of the duty cycles. It can be calculated by Eq. 4

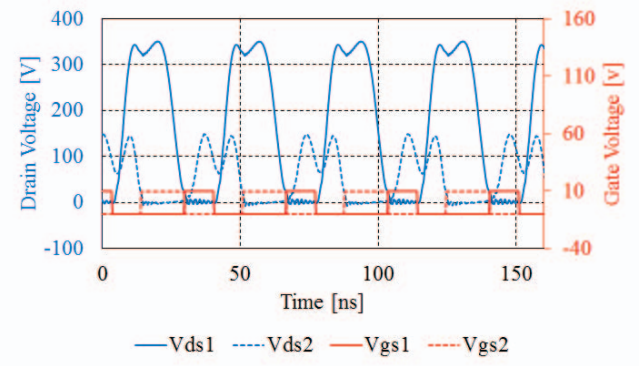
$$T_{delay} = (D_{inv} - D_{rec}) \times T_s + \frac{\Theta}{360} \times T_s \tag{4}$$

In the implemented prototype, T_{delay} = ((0.3−0.44) + (−115°/360°)) × T_s = −0.45T_s. The on-time of Q_2 is approximately 0.44, such that $v_{gs2}(t)$ turns off right before Q_1 turns on. Equation 4 gives a reference value, but the optimal value can be found through iteration.

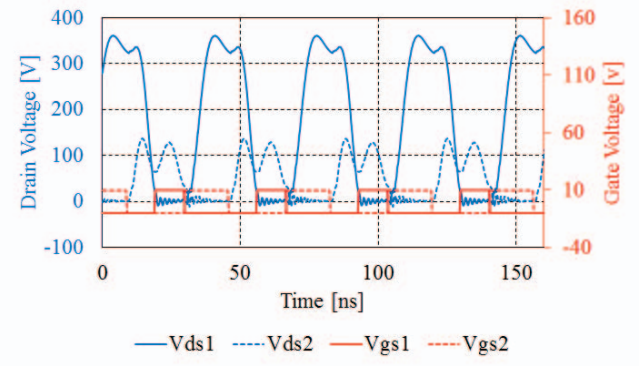
The complete dc-dc converter was simulated to verify that power could be transferred between a 170 V and a 50 V source in both directions. The effective T_{delay}s for both directions of power flow were tuned to maximize the efficiency. Drain-to-source voltages and gate-to-source voltages waveforms are shown in Fig. 5a and Fig. 5b respectively.

The gate driver was implemented using two Texas Instruments LM5114 ICs, shown in Fig. 7. The gate drive was powered by an auxiliary 7 V power supply. For simplicity, the delay between two gate signals was achieved with a simple $R - C$ delay circuit and tuned to match the simulation.

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(a) Step-down mode



(b) Step-up mode

Fig. 5: Simulation drain and gate voltage waveforms

IV. Experimental Results

Fig. 6 shows the prototype dc-dc converter and Fig. 7 shows the synchronous gate driver. Fig. 8 shows the test setup when the converter is operating in the step-down direction. Operating in a 170 V to 50 V step-down direction, the converter delivers 420 W power with an efficiency of 81 %. The converter was tested with two 2.2 Ω HSC100-2R2 wirewound power resistors in series with 1.5 Ω power resistor. The 1.5 Ω power resistor was formed by paralleling two 3 Ω HSC100-3R0 power resistors. Each of the devices is nominally rated for 100 W. The total resistance is then (2.2+2.2+3||3) = 5.9 Ω. In the step-up direction, the converter is able to boost 50 V to 160V and deliver 256 W with an efficiency of 81 %. The converter in boost direction was tested using four HSC100-100R wirewound power resistors. Each of resistor is 100 Ω rated for 100 W. Connecting two 100 Ω in parallel and followed by two parallel groups in series we made a 100 Ω load. The measurements were done with the converter operating under burst mode [2], [24], [25] with a ON/OFF frequency of 100 Hz and a duty cycle of 10 %.

The converter was also tested at full power with a nominal input voltage of 170 V. Under this condition, V_{OUT} = 49 V and P_{OUT} = 400 W with a converter efficiency of η = 78 %. The prototype converter is 0.6 inch thick and has an estimated power density of 120 W/inch^3. A simulated loss breakdown of the converter operating at nominal conditions is shown in Fig 9. The simulation assumes each inductor has a Q of 108 based on the measured Q of the inductors listed in Table III.

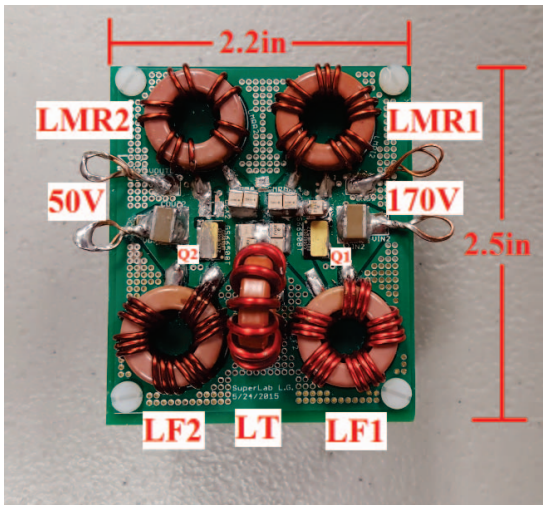


Fig. 6: Prototype bidirectional dc-dc converter, 0.6 inch thick

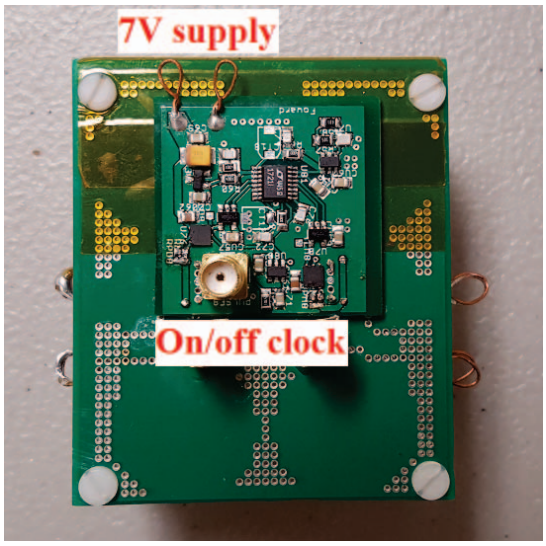


Fig. 7: Synchronous gate driver

From the loss breakdown figure, we can see the majority of the losses in the converter occur in the air-core inductors.

TABLE III: Measured Q of the inductors of the prototype converter

Inductor	Inductance [nH]	Q
L_{F1}	101	109
L_{MR1}	207	108
L_T	108	113
L_{MR2}	136.8	107
L_{F2}	46	105

The drain voltage waveforms of the active switches Q_1 and Q_2 were measured using Agilent N2875A voltage probes and capacitor divider. Capacitor divider was necessary because the probe capacitance can significantly affect the operation of the converter if used directly. Fig. 10a and Fig. 10b show the voltage waveforms at the drain nodes of the two MOSFETS.

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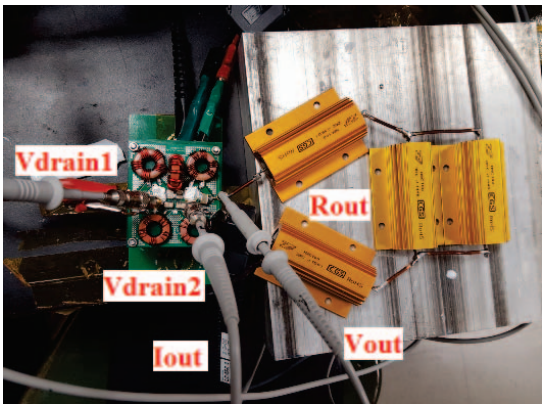


Fig. 8: Test setup for step-down direction

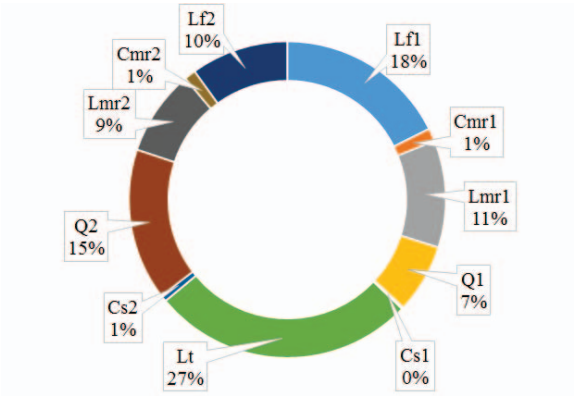
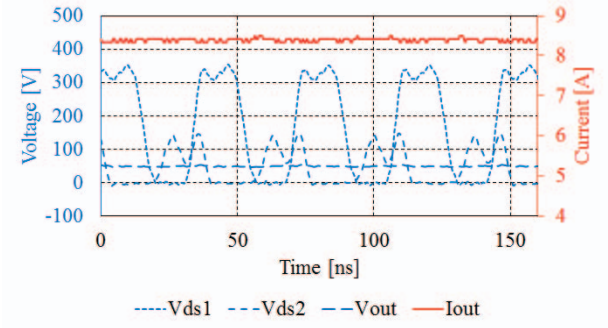
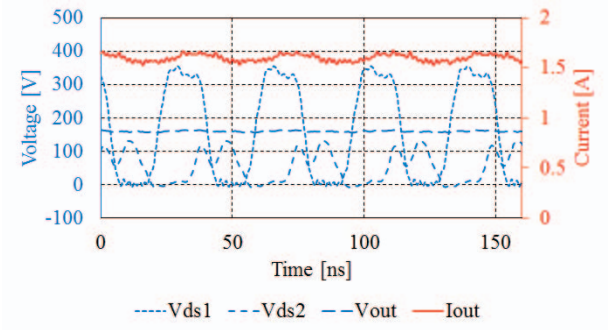


Fig. 9: Loss breakdown, V_{IN} = 170 V, V_{OUT} = 50 V, Q_L = 108



(a) Step-down direction, V_{IN}=170 V, V_{OUT}=50 V



(b) Step-up direction, V_{IN}=50 V, V_{OUT}=160 V

Fig. 10: Drain voltage waveforms, measured through capacitor divider

The waveforms of Fig. 10a were captured when the converter is operating as a down-converter and delivering power from an 170 V input to to a 50 V output. The waveforms of Fig. 10b are of the up-converter mode with an input of 50 V and output of 160 V.

Additionally, in the step-down mode of operation, the converter was tested under various input voltage conditions in the range 135 V $\leq V_{IN} \leq$ 280 V. During the test, the load was a 5.9 Ω power resistor and the converter was operated in a burst mode, with an ON/OFF frequency of 100 Hz and a modulation duty cycle of 10 %. The output voltage is not regulated at this time as the converter was running in open loop for this test. The output power and the efficiency versus the input voltage plot is shown in Fig. 12a. In the step-up direction, the converter was also tested by varying the input voltage in the range 35 V $\leq V_{IN} \leq$ 70 V. During the test, the load resistance was a 100 Ω power resistor and the converter was also operating under similar burst mode conditions. The output power and the efficiency versus the input voltage plot is shown in Fig. 12b.

V. Conclusion

This paper presents a high frequency bidirectional resonant dc-dc converter circuit based on Φ2 inverter/rectifier. A 27.12 MHz prototype converter using GaN FETs bidirectionally transferring power between 170 Vdc and 50 Vdc was implemented and demonstrated. In the step-down mode, the converter delivers 420 W from a 170 V source to 50 V load with an efficiency of 81 %. In the step-up mode, the input is 50 V while the output is 160 V. In this mode, the converter delivers 256 W with an efficiency of 81 %. The prototype dc-dc converter has a power density ≈120 W/inch^3. The majority of the power losses are in the aircore inductors. Future work will explore ways to improve the inductors to reduce losses, and to improve the power density by embedding the passive components into the PCB.

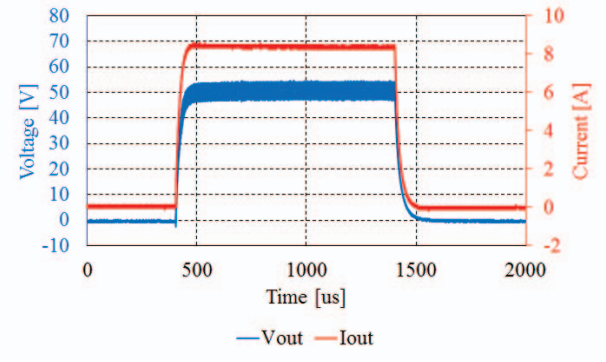
A simple way to implement the gate drive for bidirectional operation was described. Other voltage/current detection based synchronous rectifier driving scheme, such as ZVS switch on-time control driver [21] and V_{DS} active phase compensation driving scheme [7] can be exploited to further improve the performance of this high frequency synchronous resonant dc-dc converter.

This bidirectional dc-dc converter implementation circuit can find application in some dc-dc stage of offline power supply with PFC, wireless power transfer, fuel cell and battery applications, bus converters [9], [13]–[17], [21], etc. The single inverter/rectifer approach described here can be extended to multi phase or mutli level structure to accommodate different control or regulation schemes such as phase shift control, resistance compression network [22]–[24].

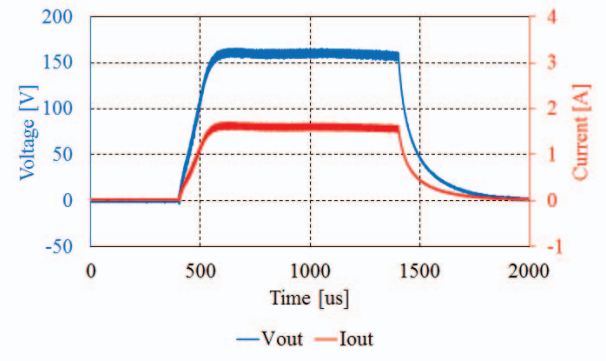
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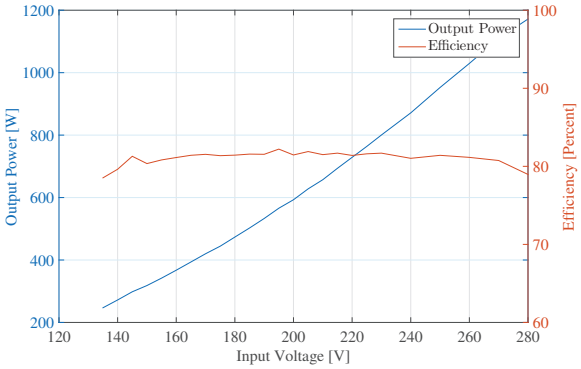


(a) Step-down direction, V_{IN}=170 V, V_{OUT}=50 V

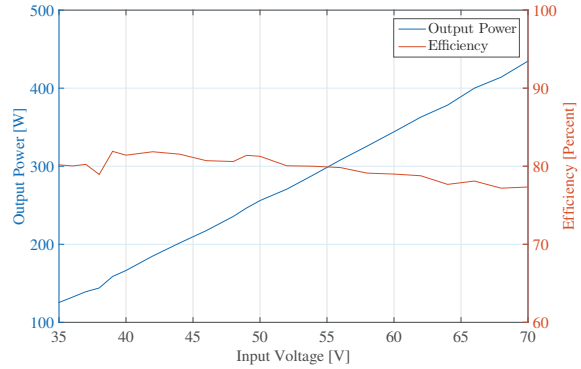


(b) Step-up direction, V_{IN}=50 V, V_{OUT}=160 V

Fig. 11: Measured output transient waveforms



(a) Step-down direction



(b) Step-up direction

Fig. 12: V_{OUT} unregulated, 100 Hz 10 % burst mode

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